

EVENT PROPOSAL

EDGE AI CHALLENGE

“Train it. Optimize it. Deploy it on the edge.”

1. Overview

Event Type	Team Technical Competition
Domain	TinyML, Edge Computing & Embedded AI
Competition Duration	2 Hours
Participation	Teams of 2–3 members
Evaluation Platform	Standardized organizer-controlled microcontroller

Concept: Participants receive a common Edge AI problem statement, dataset, target-platform specifications and submission requirements at the start of the competition. Teams use the two-hour competition period to develop, train and optimize a lightweight machine-learning model suitable for embedded deployment.

At the submission deadline, each team submits its finalized model and deployment package. The organizing team subsequently deploys and benchmarks all valid submissions on a standardized microcontroller platform under identical test conditions. Final ranking is based on model performance, inference latency, resource usage and successful on-device execution.

2. Objectives

- Introduce participants to practical Edge AI and TinyML development workflows.
- Test the ability to design machine-learning solutions for resource-constrained hardware.
- Encourage optimization of model performance, inference speed and memory usage.
- Provide exposure to preparing trained models for embedded deployment.
- Highlight the engineering trade-offs involved in running AI inference directly at the edge.

3. Competition Format

Official competition period: 2 hours

At the start of the timed round, all teams receive:

- The official problem statement and dataset.
- Target hardware specifications and model constraints.
- Evaluation criteria and required submission format.
- A deployment template or interface specification, where required.

During the two-hour development period, teams may:

- Preprocess and prepare the supplied data.
- Train and validate a machine-learning model.
- Apply quantization, compression or other suitable optimization techniques.
- Convert the model into the required embedded-compatible format.
- Prepare the final deployment files and submission documentation.

At the end of the two-hour period, submissions are frozen. No model or code changes are permitted after the official deadline.

4. Post-Event Organizer Evaluation

Following the competition period, the organizing team evaluates all valid submissions on the designated embedded platform. Each submission is tested using the same hardware configuration, deployment procedure and hidden evaluation inputs to maintain consistent benchmarking across teams.

The evaluation includes:

- Verification that the submitted model successfully executes on the target microcontroller.
- Model performance on organizer-provided hidden evaluation data or inputs.
- Measurement of on-device inference latency.
- Measurement of model size and relevant embedded resource usage.
- Review of the submitted technical approach and optimization choices.

Organizers will evaluate the submitted files as received. Organizer-side changes will be limited to the standard steps required to build, flash and benchmark the submission using the published deployment interface.

5. Judging & Marking Scheme

Total: 100 points

Criterion	Weight	Evaluation
Model Performance / Accuracy	45 pts	Performance of the final model on organizer-provided hidden evaluation data or inputs.
Inference Latency	20 pts	Time required by the target microcontroller to execute inference.
Model Size / Resource Usage	15 pts	Efficiency of the submitted model with respect to model size and available embedded resources.
Successful Edge Deployment	10 pts	Correct execution of the submitted solution on the target microcontroller.
Technical Approach & Documentation	10 pts	Clarity of the submitted approach, model choice, preprocessing and optimization decisions.

6. Event Flow

A short rules briefing may be completed before the official two-hour timer begins so that the full competition period is available for model development.

Stage / Time	Activity
Before Timer	Registration confirmation, participant setup and rules briefing
0:00	Problem statement, dataset and submission requirements released; official timer begins
0:00 – 2:00	Model development, training, validation, optimization and deployment preparation
2:00	Final submission deadline and submission freeze
Post-Event	Organizer-led deployment, hidden-data evaluation and hardware benchmarking
After Evaluation	Score verification and results announcement

7. Submission Requirements

Each team must submit the following before the official deadline:

- Final trained/converted model in the specified format.
- Required deployment source code or project files.
- Model configuration information required for evaluation.

- Model size and basic validation results.
- A brief technical note describing the model, preprocessing and optimization approach.
- Team details.

The version submitted at the deadline is the version used for final evaluation.

8. Rules & Fair Evaluation

- All teams receive the official problem statement and dataset at the same time.
- Teams must follow the published target-platform constraints and submission format.
- Submissions received after the deadline will not be accepted unless an organizer-side issue has been documented.
- Final scored inference must execute locally on the designated embedded platform.
- Cloud-based or remote inference is not permitted during organizer benchmarking.
- Hard-coded, prerecorded or manually substituted prediction outputs are prohibited.
- Participants must not copy another competing team's model, code or solution.
- Organizers may inspect submitted code, model files and runtime behavior to verify compliance.
- All teams will be benchmarked using the same evaluation procedure and target hardware configuration.

9. Hardware & Software Requirements

Organizer Setup

- Standardized target microcontroller/development board for post-event evaluation.
- Organizer laptop with the required board drivers, libraries and benchmarking environment.
- Tested deployment template/interface and build procedure.
- Training/validation dataset and separate hidden evaluation set.
- Scoring sheet and standardized benchmark procedure.
- USB/power accessories and any required peripherals for the target platform.
- Display/projector for briefing or result announcement, if available.

Participant Setup

- Personal laptop and charger.
- Required programming and machine-learning development environment as communicated by organizers.
- Any permitted local tools required for model training, conversion and submission preparation.

Target-platform specifications and required software should be communicated in advance so participants can complete environment setup before the event.

10. Tie-Breaking

If two or more teams receive the same final score, ranking will be determined in the following order:

- Higher model-performance score.
- Lower inference latency.
- Smaller model/resource usage.
- Higher technical-approach score.

11. Expected Outcome

The Edge AI Challenge gives participants practical exposure to the workflow of preparing an AI model for resource-constrained embedded execution:

Problem & Dataset → Model Development → Optimization → Submission → Embedded Deployment → On-Device Benchmarking

The event emphasizes a central requirement of Edge AI: a successful solution must not only achieve strong model performance, but must also be compact, responsive and capable of executing directly on constrained embedded hardware.